MILITARY SPECIFICATION

REGULATOR KIT, HOSELINE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a hoseline regulator kit for a petroleum hoseline.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3835

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MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-12044	- Chests, Plywood.
MIL-P-15024/6	- Plates, Identification, Equipment.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

A2 13225E9046	- Cable Assembly, Regulator Kit.
A2 13225E9047	- Hose Assembly, Pressure Gage.
A2 13225E9048	- Bracket, Mounting, Control Box.
A3 13225E9049	- Block, Insulation.
A2 13225E9050	- Regulator Kit, Hoseline.
A3 13225E9051	- Spacer.
A2 13225E9052	- Cable assembly, Regulator Kit.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Description. The hoseline regulator shall be furnished as a complete kit. Construction details of the individual items in the kit shall be in accordance with Drawings A2 13225E9046 through A2 13225E9048, A3 13225E9049, A2 13225E9050, A3 13225E9051, and A2 13225E9052, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the hoseline regulator. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. The material shall be as specified herein and on the applicable drawings. However, when a definite material is not specified, a material shall be used which will enable the hoseline regulator to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product (see 6.4).

3.4 Threads. All screw threads shall conform to FED-STD-H28.

3.5 Pump pressure regulating assembly. The contractor shall furnish a pump pressure regulating assembly in accordance with Drawing A2 13225E9050 to be compatible with the Deutz engine model F3L912 and the Gorman-Rupp pump model 04A1.

3.5.1 Operation. A three-position rotary switch provides selection of the desired mode of operation: START-AUTO-MANUAL. In the MANUAL mode, automatic pressure control is not provided. The pump speed is isochronously controlled; that is, the engine-pump speed stays at the desired set speed, irrespective of pump pressure, up to the maximum power output of the engine. The speed is set by the Manual Speed Control knob located on the face of the pressure regulating control. The adjustable speed range is from 800 revolutions per minute (rpm) to 2450 rpm.

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3.5.1.1 Start mode. In the START position the engine is commanded to run at 800 rpm. An interlock to the engine start circuit prevents starting this engine except in this position. The Manual Speed Control has no effect. The purpose of this feature is to prevent accidentally operating the engine and pump at a high speed without liquid in the pump or without a brief warm-up time for the engine.

3.5.1.2 Auto mode. In the AUTO mode, the pressure regulating control shall operate into an electronic speed control governor which has proportional, integral and derivative action to control the discharge pressure of the pump to 125 ±5 pounds per square inch gage (psig). The system will maintain a steady state discharge pressure from minimum to maximum flow conditions while the pump is being supplied with a suction pressure of more than 20 psig. If the suction pressure to the pump drops below 20 psig, the pressure regulator will proportionally decrease the engine speed down to an idle at 10 psig or below to prevent collapsing the suction hose. The pressure sensing and speed command components are all solid state electronics. They operate an electro-mechanical actuator to precisely position the fuel control on the engine for isochronous (zero drop) speed setting.

3.5.2 Hydrostatic pressure. The regulator shall withstand an internal hydrostatic pressure of not less than 300 psig applied at the discharge connections and not less than 150 psig applied at the suction connections.

3.6 Chest. All components of the hoseline regulator shall be furnished in a plywood chest conforming to MIL-C-12044. Partitions shall be provided for components of the hoseline regulator to prevent damage during shipment or handling.

3.7 Treatment and painting. The portions of the hoseline regulator normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. Nonpermissible galvanic couples (dissimilar metals) as defined in MIL-T-704 shall be insulated as specified in MIL-T-704.

3.8 Identification marking. The hoseline regulator shall be identified in accordance with MIL-STD-130. The marking shall be applied to the hoseline regulator on plates conforming to MIL-P-15024/6, type H, style III.

3.9 Workmanship. All components of the hoseline regulator shall be clean and free from rust, scale, flux, and other extraneous material. All components shall fit into the chest without binding or forcing. Burrs, nonfunctional sharp edges, projecting points and excessive length of fastening devices shall be avoided.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows.

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article hoseline regulator shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article hoseline regulator shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each hoseline regulator shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective.

4.4.2 Tests. Each hoseline regulator shall be tested as specified in 4.5.2. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The hoseline regulator shall be examined as specified herein for the following defects:

101. Dimensions not as shown on the drawings.
102. Materials not as specified.
103. Components missing or not as specified.
104. Treatment and painting not as specified.
105. Identification marking missing, incomplete, or illegible.
106. Workmanship not as specified.

4.5.2 Test.

4.5.2.1 Pressure control system test.

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4.5.2.1.1 Assembly. Set the pressure regulator and the electronic speed governor to control the speed of an actual engine or an equivalent simulated test circuit with a speed feedback from the fuel control actuator. Connect the suction pressure sensing element to an adjustable pressure source and pressurize the port to 30 psig. Connect the discharge pressure sensing element to an adjustable pressure source and pressurize the port to 100 psig. Attach the electrical connections to a 12 volt direct current source.

4.5.2.1.2 Manual speed control. Place the mode selector switch in the MANUAL position. Demonstrate that with the Manual Speed Control adjusted to IDLE position, the engine speed is 800 rpm ±20 rpm or equivalent speed signal from a test fixture. Turn the Manual Speed Control to the HIGH position and demonstrate that the engine speed is 2450 rpm ±75 rpm or equivalent.

4.5.2.1.3 Discharge pressure check. Place the mode selector switch in the AUTO position. With pressures adjusted to 100 psig at the discharge port and 30 psig at the suction port the engine speed shall be 2450 rpm ±75 rpm or equivalent. Gradually increase the pressure to the discharge port to 135 psig. The engine speed shall reduce and settle out at 800 rpm ±20 rpm at 125 ±5 psig. Reduce the pressure to the discharge port to 118 psig ±2 psig. The engine speed shall increase to 2450 rpm ±75 rpm.

4.5.2.1.4 Suction pressure check. With the mode selector in the AUTO position and 100 psig to the discharge port and 30 psig to the suction port, the engine shall be operating at 2450 rpm ±75 rpm. Gradually lower the pressure to the suction port. The engine speed shall behave as follows:

a. At 20 psig ±1.5 psig, the speed shall start to reduce from 2450 rpm ±75 rpm.

b. At 10 psig ±1.5 psig and below the engine speed shall be at 800 rpm ±30 rpm.

c. Increase the suction pressure to 21.5 psig. The engine speed shall return to 2450 rpm ±75 rpm. The engine shall go through its full speed range of 800 rpm ±30 rpm to 2450 rpm ±75 rpm in a proportional manner as the suction pressure changes from 10 psig ±1.5 psig to 20 psig ±1.5 psig and vice versa.

4.5.2.1.5 Failure criteria. Failure of the engine speed to respond as described in test procedure above, after adjustments have been made, shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

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4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Materials, methods, and containers not as specified for level A or B. Each incorrect material, method, or container shall be considered one defect.
108.	Components not cushioned and secured within the chest as specified for level A or commercial.
109.	Components not preserved and packed in accordance with the referenced document, for commercial.
110.	Chest not blocked within box as specified for level A or B.
111.	Marking, missing, illegible, incorrect or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. The kit components shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3 and placed, cushioned, and secured within the chest.

5.1.2 Commercial. The kit components shall be preserved in accordance with MIL-STD-1188, and placed, cushioned and secured within the chest.

5.2 Packing. Packing shall be level A, B, or commercial as specified, (see 6.2).

5.2.1 Level A. The chest, containing the kit components, shall be packed in a close-fitting box conforming to PPP-B-636, V3c, style optional. The chest shall be blocked within the box with corrugated fiberboard pads to prevent movement and damage. Box closure shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Packing shall be as specified in 5.2.1, except the box shall be class domestic, grade 200.

5.2.3 Commercial. The chest containing the kit components, shall be packed in accordance with MIL-STD-1188.

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5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The hoseline regulator is intended for use in manual and automatic modes of engine speed regulation and automatic pump output regulation on the 350 gallons per minute pump, MIL-P-52144.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.

(b) When a first article is required for inspection and approval (see 3.2, 4.3 and 6.3).

(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item will be tested and should be a sample selected from the first five production items. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations and test approval of the document's first article.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-R-53001(ME) Regulator Kit, Hoseline

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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